Expert Information

Expert Information

First Name

Last Name

Email

Position Title

Organization / University

Instructions

Purpose

The purpose of this research is to develop and validate the content for a new **Quantum Information Science Concept Introductory Test (QISCIT)**.

As implied by its name, QISCIT focuses on introductory QIS concepts that can be demonstrated without the formalism or linear algebra required by other quantum assessments. QISCIT has been designed to focus on the following quantum concepts: *qubits*, *gates*, *entanglement*, *teleportation*, and *decoherence*.

We are employing the process of Almanasreh et al. (https://doi.org/10.1016/j.sapharm.2018.03.066) to evaluate the content validity of QISCIT. This process consists of two rounds: 1) quantifying the content validity of each item and the initial overall instrument based on ratings from experts, such as yourself, to determine which items to delete, modify, or replace, and 2) quantifying the content validity of each revised item and the final overall instrument. Your estimated commitment is a total of 2 hours: 1 hour for the first round and 1 hour for the second round. **This is the second round of the process.**

The development and validation of QISCIT is funded by the National Science Foundation (NSF) Award #2302816: "Collaborative Research: Advancing Quantum Education by Adaptively Addressing Misconceptions in Virtual Reality".

Instructions

On the following pages, you will be presented with the **second version of QISCIT**, which includes a glossary of QIS concepts and **a total of 34 questions** grouped by primary concept for your convenience. For each item, we ask that you **rate** the item based on content relevance from 1 ("Not Relevant") to 4 ("Extremely Relevant") and **provide comments** and make suggestions regarding the wording of the item.

Note, the groups of questions will be randomized to mitigate potential learning effects due to ordering.

Glossary of Quantum Information Science Concepts

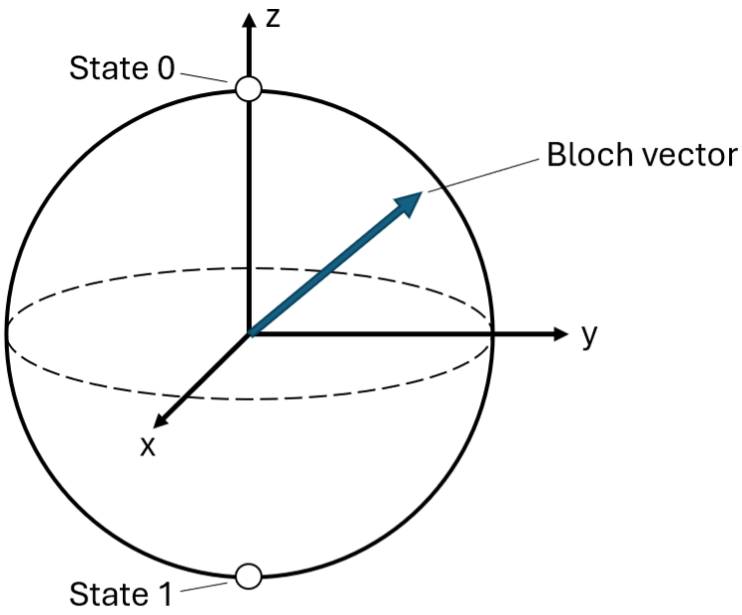
Note, there have been several edits to the glossary, so please review it carefully.

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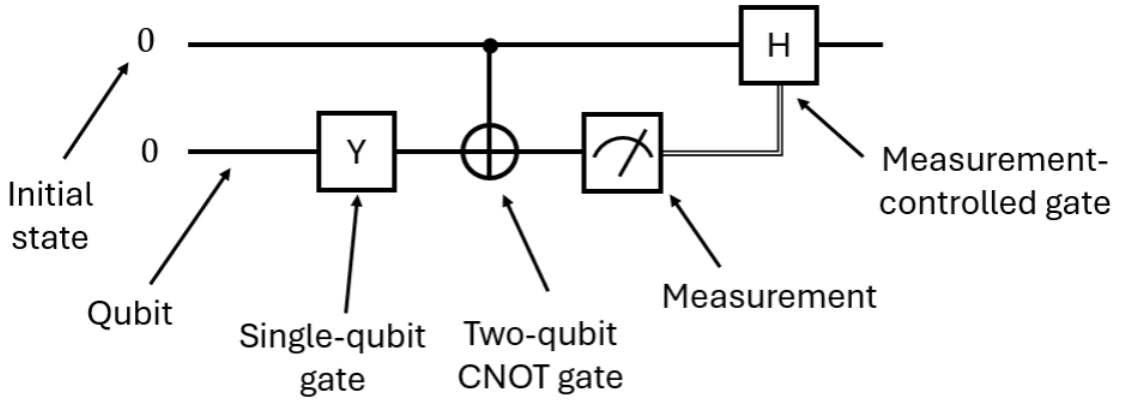
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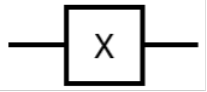
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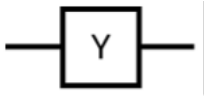
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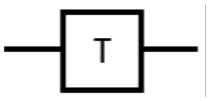
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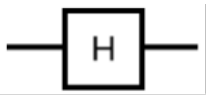
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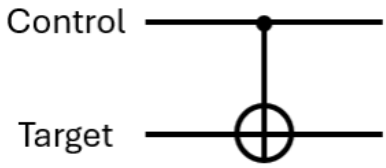
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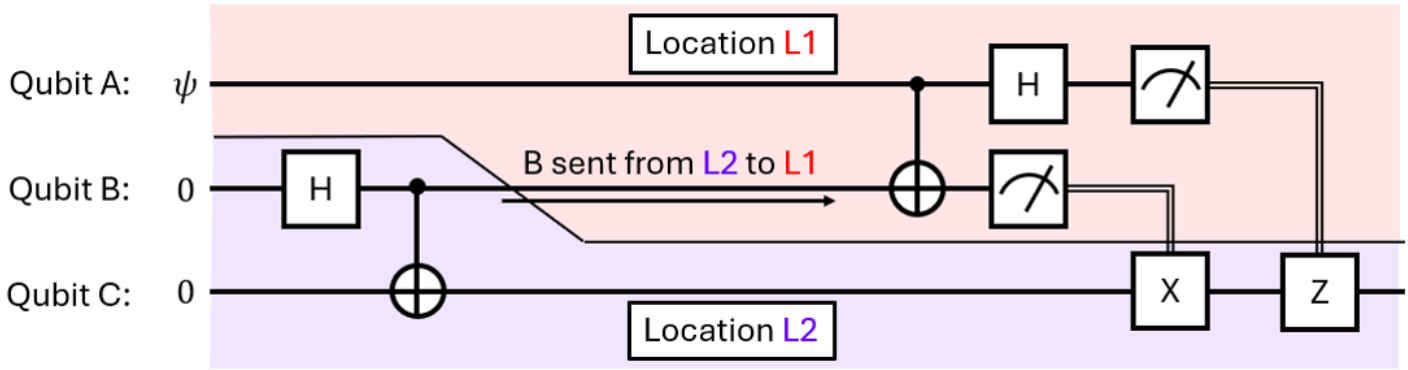


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Please rate the glossary above based on content relevance.

1 - Not Relevant	2	3	4 - Extremely Relevant

Please provide any comments or suggestions regarding the wording of the glossary above.

QIS Concept Introductory Test

Please answer the following questions to the best of your ability using your knowledge of quantum information science and the glossary above.

The following question is intended to serve as an attention check for data analysis purposes.
How many degrees does the T gate rotate a qubit's Bloch vector around the Z axis?

45 degrees

90 degrees

180 degrees

The T gate does not rotate around the Z axis

Please rate the item above based on content relevance.

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Qubits

Qubits

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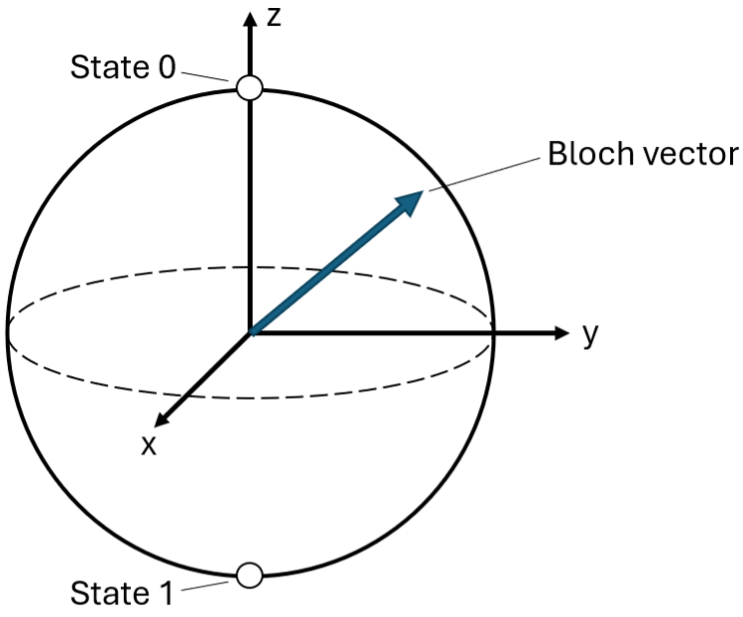
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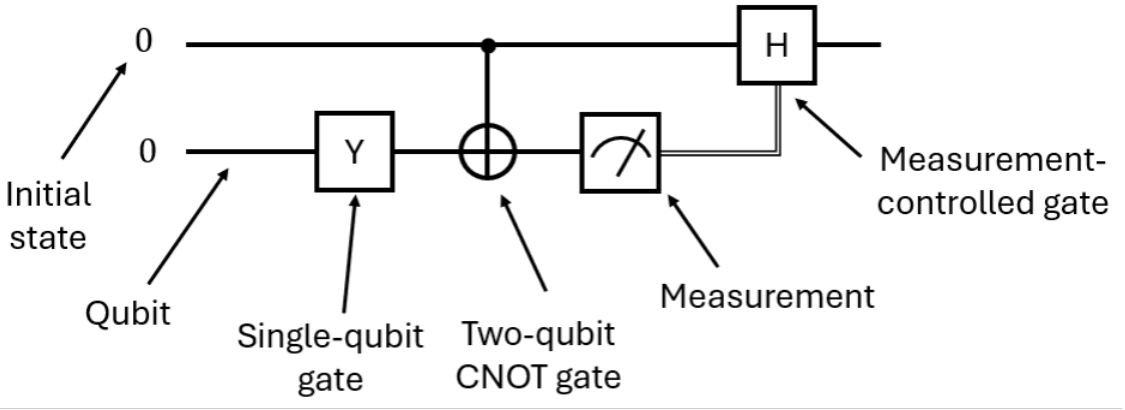
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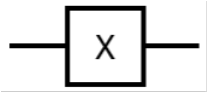
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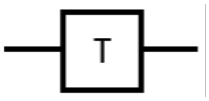
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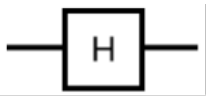
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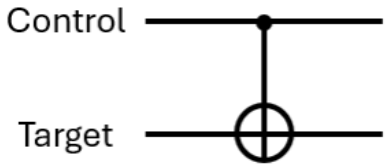
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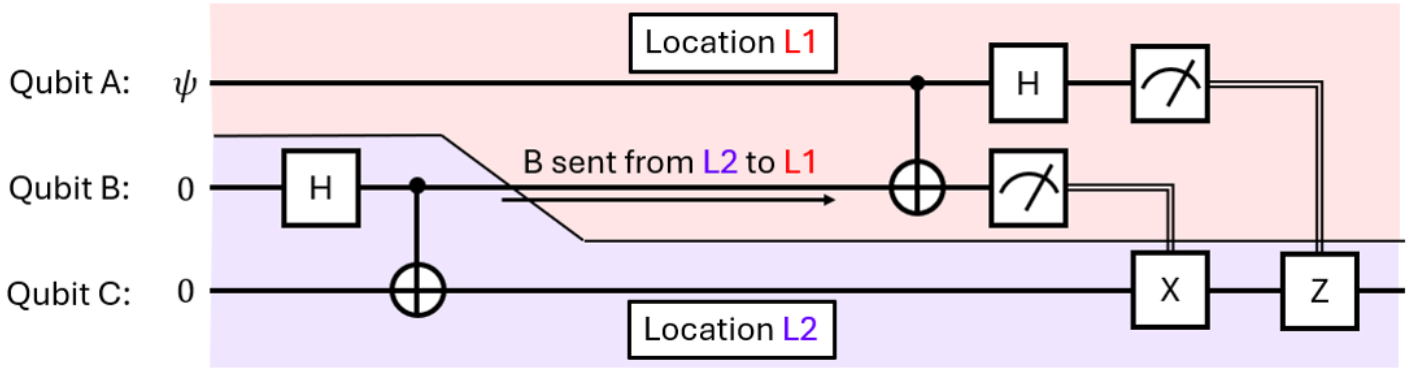
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Provide any additional comments or suggestions regarding the wording of the glossary here.

Question

A single isolated qubit is in a superposition of states 0 and 1. What quantum state will it be in after a measurement?

The qubit will be in both state 0 and state 1

The qubit will be in either state 0 or state 1

The qubit will be in neither state 0 nor state 1

The qubit will remain in state 0 or state 1

Please rate the item above based on content relevance.

1 - Not Relevant 2 3 4 - Extremely Relevant

Please provide any comments or suggestions regarding the wording of the item above.

Question

For a single qubit, how many states exist that are orthogonal to the state 0?

Zero

One

Two

Infinity

Please rate the item above based on content relevance.

1 - Not Relevant 2 3 4 - Extremely Relevant

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Question

How many potential superpositions exist for a single qubit?

Zero
One
Two
Infinity

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Question

How many possible measurement outcomes does a single qubit in a superposition state have?

Zero
One
Two
Infinity

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1 - Not Relevant | 2 | 3 | 4 - Extremely Relevant

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Question

How many possible measurement outcomes does a single qubit have immediately after a prior measurement in the same basis?

Zero

One

Two

Infinity

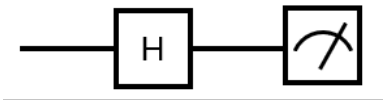
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Question

A qubit starts in an unknown superposition state. An H gate is applied, followed by a measurement. How can the qubit be returned to its original unknown state?



Apply another H gate and then measure again

Apply another H gate but do not measure again

Measure the qubit again and then apply another H gate

It is not possible to recover the initial unknown state

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Question

A qubit in an unknown state is measured, which returns state 0. What is known about the qubit's state prior to the measurement?

The qubit was in a superposition

The qubit was in state 0

The qubit was not in state 0

The qubit was not in state 1

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Question

How many basis states must a physical system have in order to use it as a qubit?

An infinite number

At least one

At least two

Exactly two

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Question

A single qubit starts in an unknown state. The qubit is then measured, which returns state 0. The same unknown state is then prepared and measured again. What is certain about the results of the second measurement?

It could be 0 or 1

It will definitely be 0

It will definitely be 1

It is equally likely to be 0 or 1

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Gates

Gates

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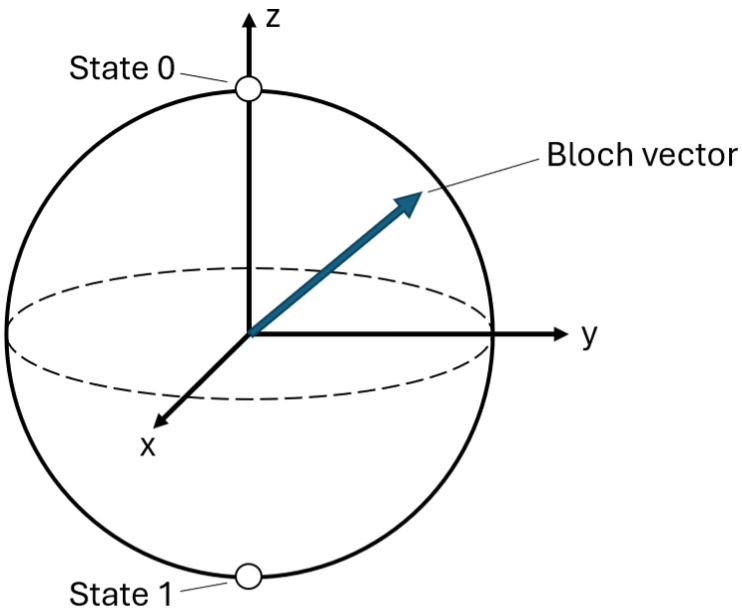
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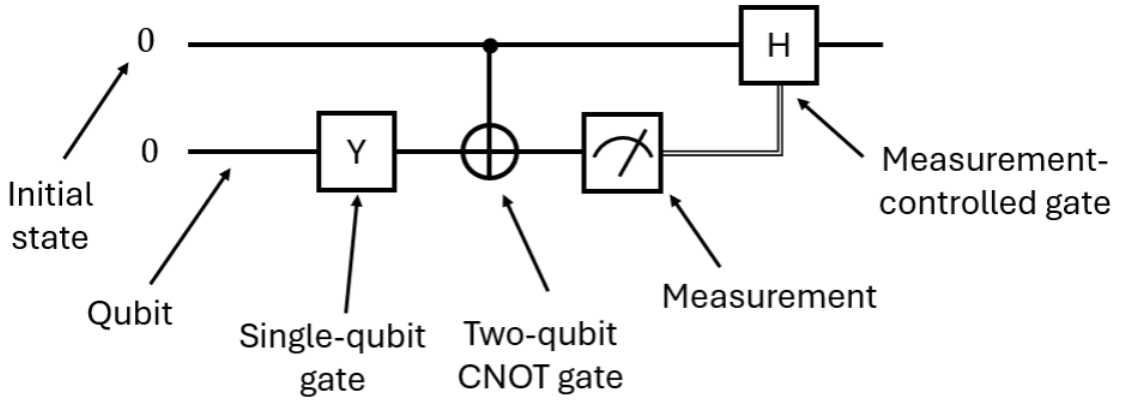
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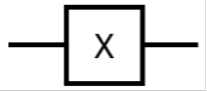
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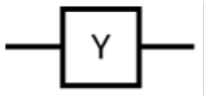
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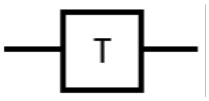
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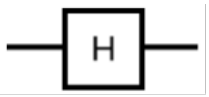
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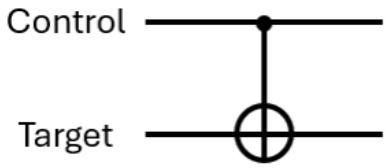
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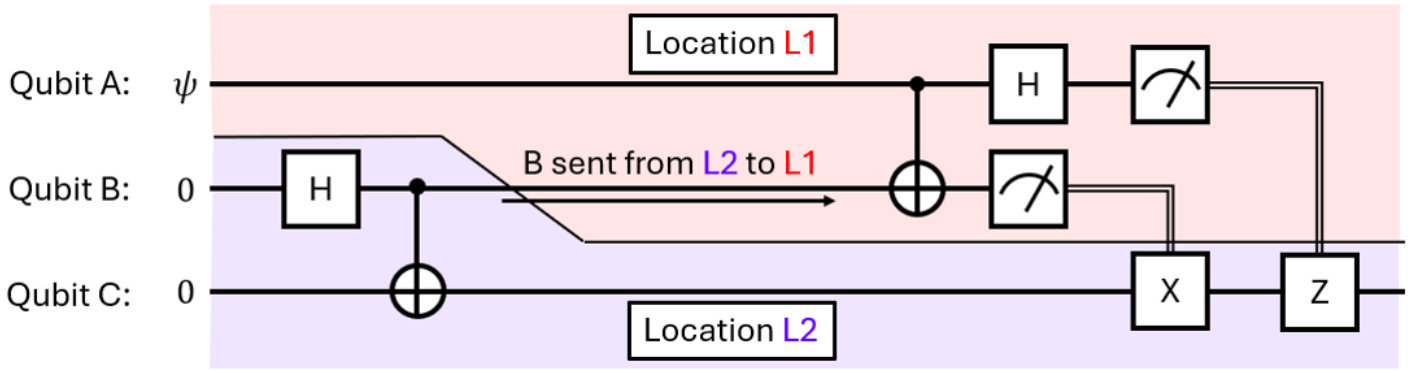


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Question

When are ideal quantum gates not reversible?

Only when the qubits are entangled

Only when the qubits are in superpositions

Quantum gates are always reversible

Quantum gates are never reversible

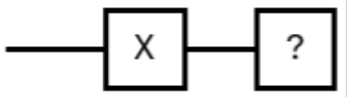
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Question

How can the operation of an X gate be reversed?



Apply an H gate

Apply an X gate

Apply a Y gate

The X gate cannot be reversed

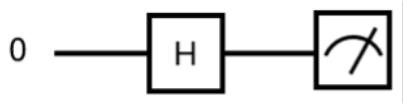
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Question

A qubit starts in state 0 and then has an H gate applied to it. If the qubit is then measured, what is the probability it will return state 0?



0%

25%

50%

75%

100%

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Question

How can a qubit be put into superposition starting from state 0?

Apply an H gate

Apply an S gate

Apply a T gate

Apply an X gate

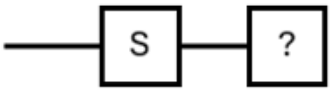
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Question

How can the operation of an S gate be reversed?



Apply another S gate

Apply another two S gates

Apply another three S gates

The S gate cannot be reversed

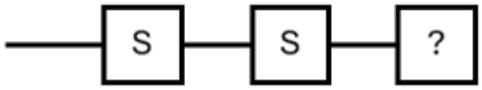
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Question

How can the operations of two S gates be reversed?



Apply one T gate

Apply two T gates

Apply four T gates

The S gates cannot be reversed

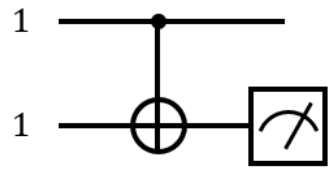
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1 - Not Relevant 2 3 4 - Extremely Relevant

Please provide any comments or suggestions regarding the wording of the item above.

Question

Two qubits each start in state 1. A CNOT gate is applied, using the first qubit as the control and the second qubit as the target. If the second qubit is then measured, what is the probability it will return state 0?



0%

25%

50%

75%

100%

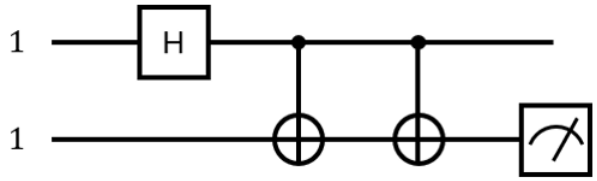
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Question

Two qubits each start in state 1. An H gate is applied to the first qubit. Two CNOT gates are then applied, using the first qubit as the control and the second qubit as the target. If the second qubit is then measured, what is the probability it will return state 0?



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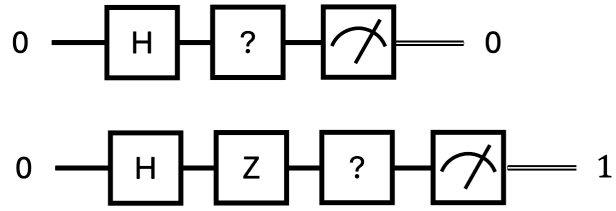
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Question

The same unknown gate is applied in the following two quantum circuits. The first quantum circuit ends with a measurement, which always returns state 0. The second

quantum circuit ends with a measurement, which always returns state 1. What common single-qubit gate could have been applied as the unknown gate?



An H gate

An X gate

A Z gate

No such gate exists

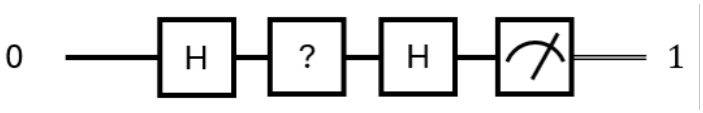
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Question

A qubit starts in state 0. An H gate, an unknown gate, and an H gate are applied to the qubit. This quantum circuit ends with a measurement, which always returns state 1. What common single-qubit gate could have been applied as the unknown gate?



An S gate

An X gate

A Y gate

An X or Y gate

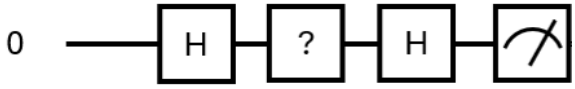
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Question

A qubit starts in state 0. An H gate, an unknown gate, and an H gate are applied to the qubit. This quantum circuit ends with a measurement, which returns state 1 with a probability more than 0% but less than 50%. What common single-qubit gate could have been applied as the unknown gate?



An S gate

A T gate

An X gate

A Y gate

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Entanglement

Entanglement

NOTE: We are repeating our glossary here for your convenience.

Glossary of Quantum Information Science Concepts

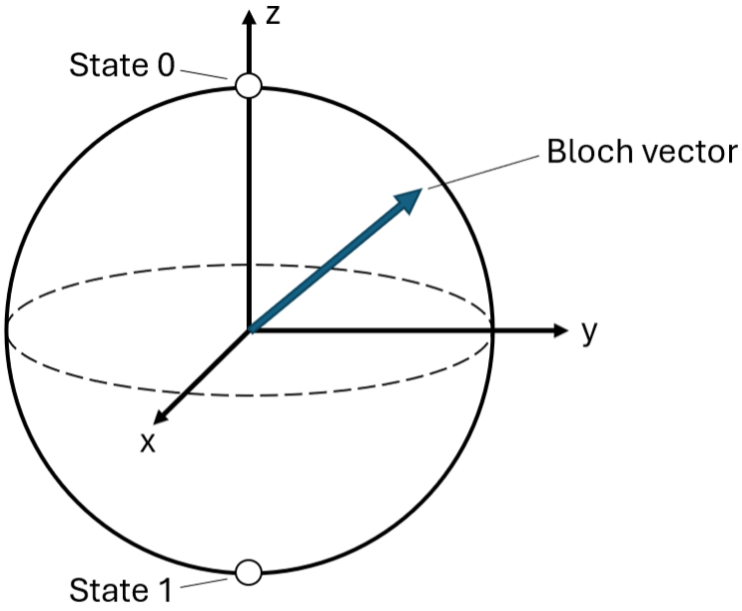
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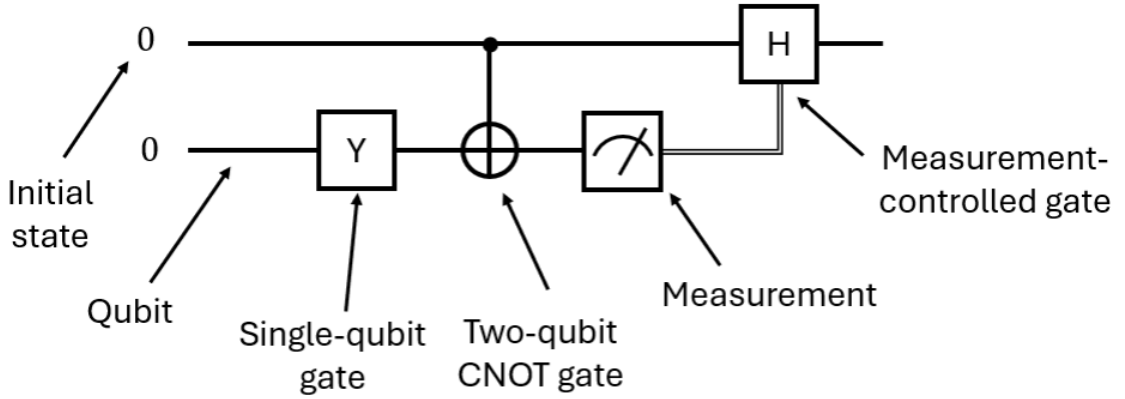
Superposition: The ability of a quantum system to be in multiple basis states at the same time.

Bloch Vector: A method of visualizing the state of a single qubit by converting information

about its quantum state to a point on or within a sphere. For more than one qubit, the Bloch vector representation does not always provide complete information. A picture of a Bloch vector is shown here.



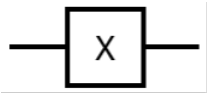
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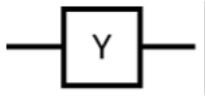
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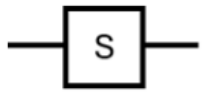
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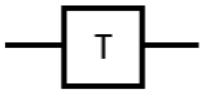
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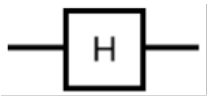
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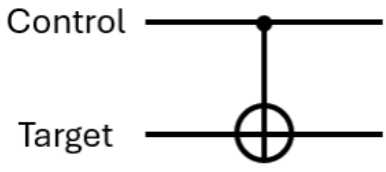
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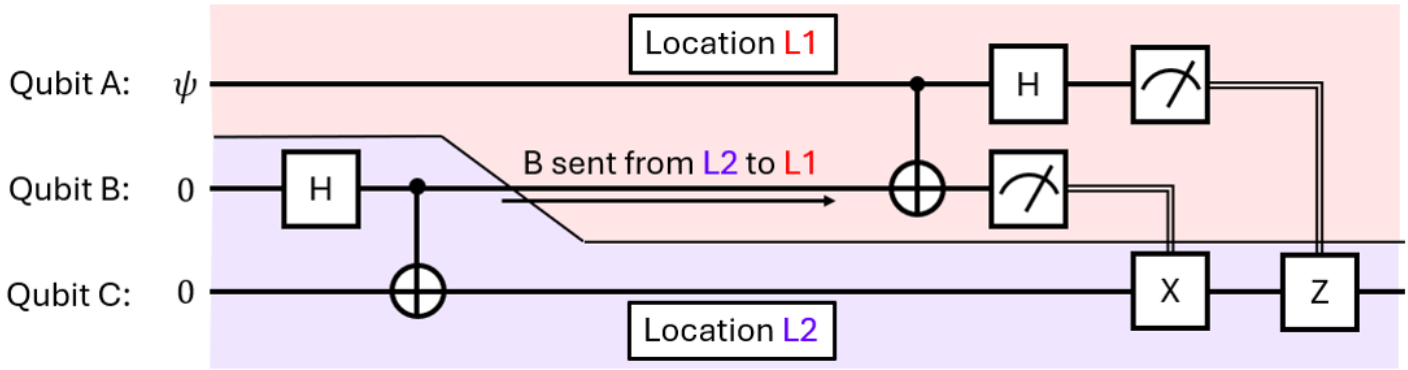
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Provide any additional comments or suggestions regarding the wording of the glossary here.

Question

Two qubits are entangled, and then the first qubit is measured. What happened to the entanglement between the two qubits?

The entanglement decreased to zero

The entanglement depends on the result of the measurement

The entanglement increased

The entanglement remained the same

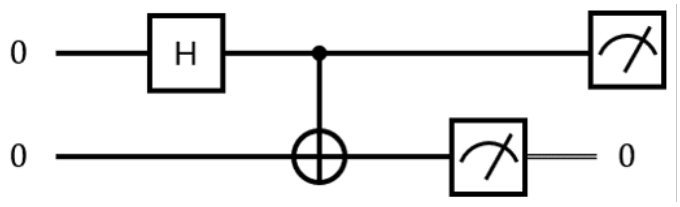
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Question

Two qubits each start in state 0. An H gate is applied to the first qubit. A CNOT gate is then applied, using the first qubit as the control and the second qubit as the target. The second qubit is measured, which returns state 0. If the first qubit is then measured, what is the probability it will return state 0?



0%

25%

50%

75%
100%

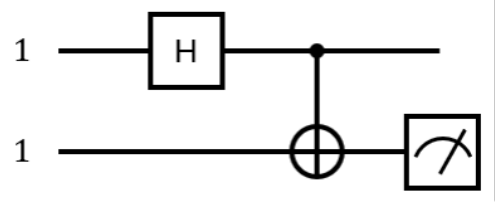
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Two qubits each start in state 1. An H gate is applied to the first qubit. A CNOT gate is then applied, using the first qubit as the control and the second qubit as the target. If the second qubit is then measured, what is the probability it will return state 0?



0%
25%
50%
75%
100%

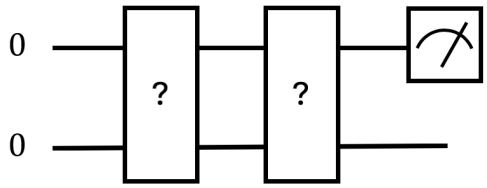
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Question

Two qubits each start in state 0. A series of unknown gates are applied to both qubits. The quantum circuit ends with a measurement of the first qubit, which returns state 0 with a probability of 50%. What is known about the first qubit before the measurement?



It is entangled and in superposition

It is entangled or in superposition

It is only entangled

It is only in superposition

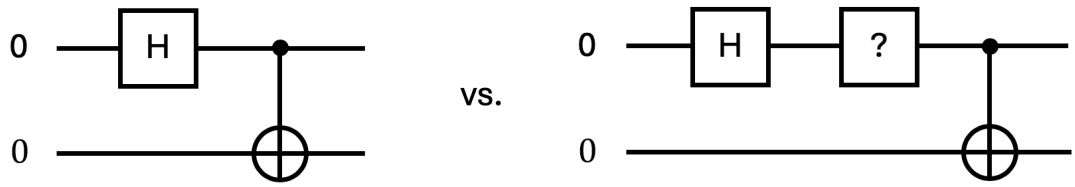
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Question

The following quantum circuits produce orthogonal Bell states. What common single-qubit gate could have been applied as the unknown gate?



An S gate

A T gate

An X gate

A Z gate

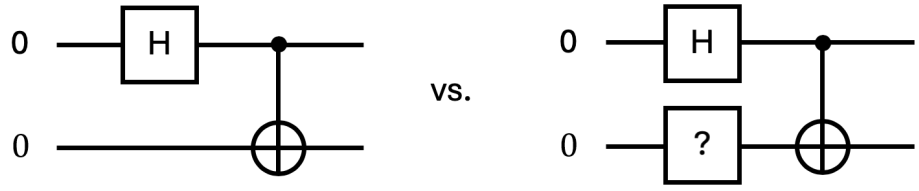
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An H gate

An S gate

A Y gate

A Z gate

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Teleportation

Teleportation

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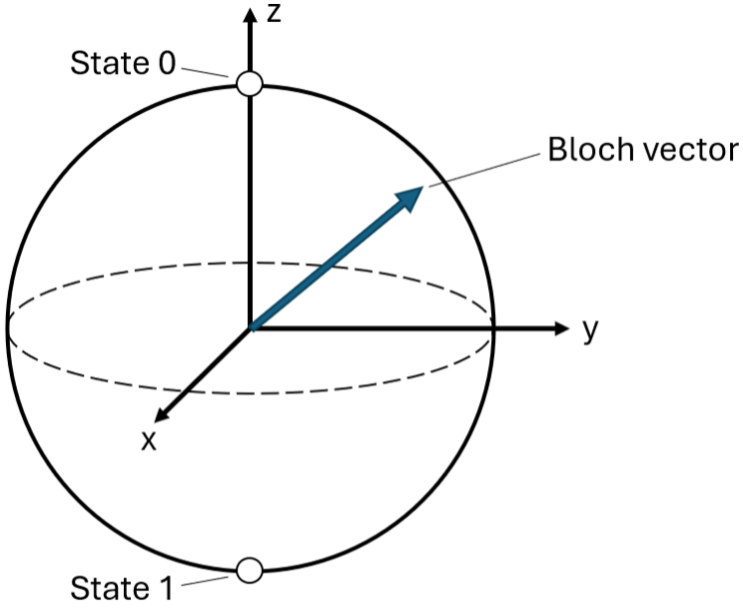
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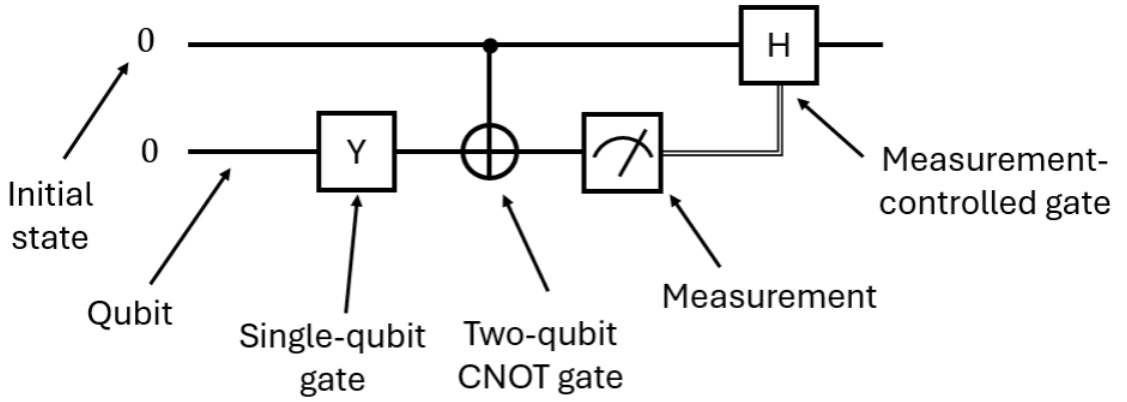
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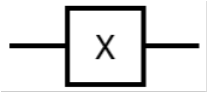
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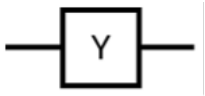
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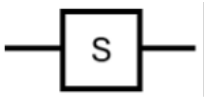
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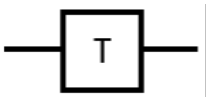
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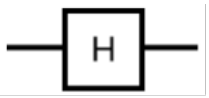
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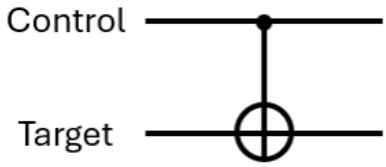
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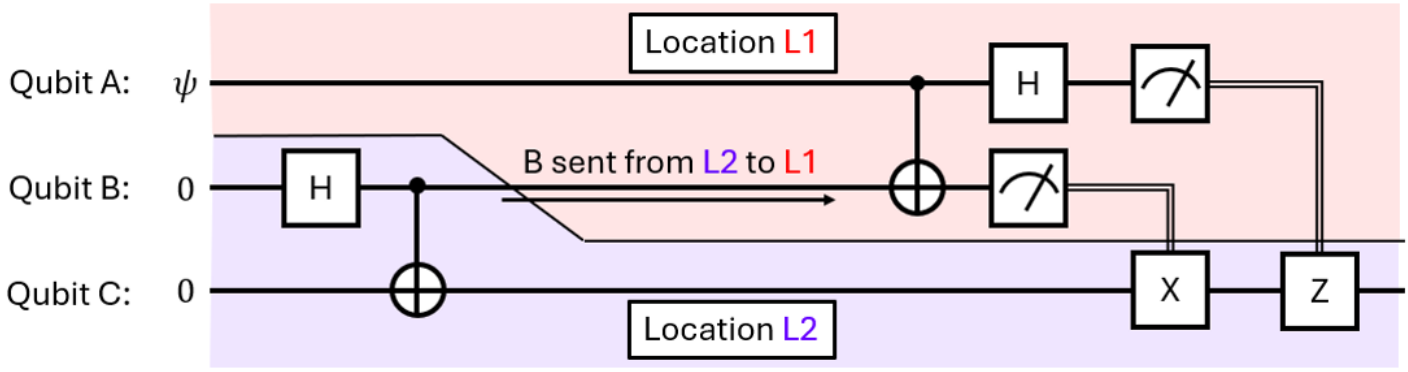
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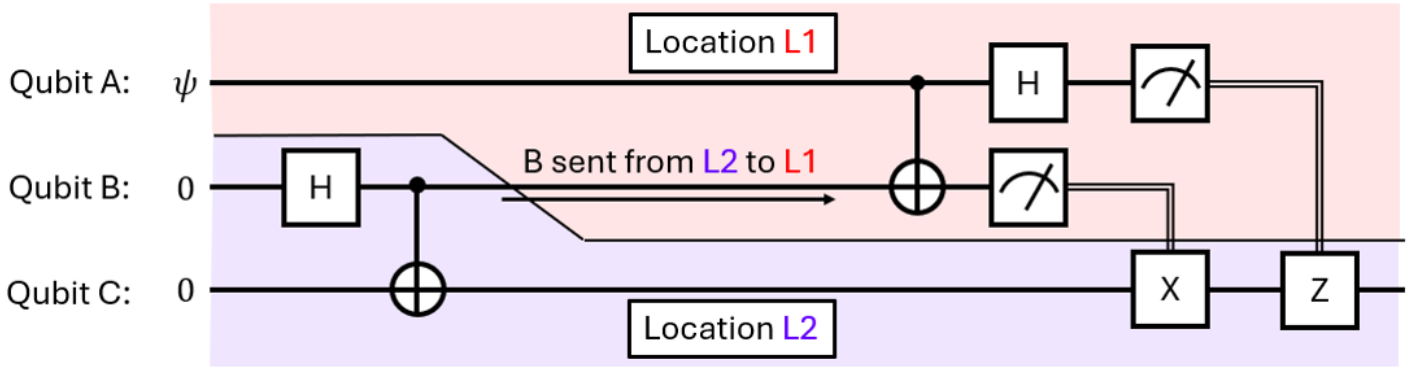


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Provide any additional comments or suggestions regarding the wording of the glossary here.

Question

What piece of quantum information does the following quantum teleportation circuit transfer from location L1 to location L2?



Any Bell state

Any single-qubit state

Any two-qubit state

Any three-qubit state

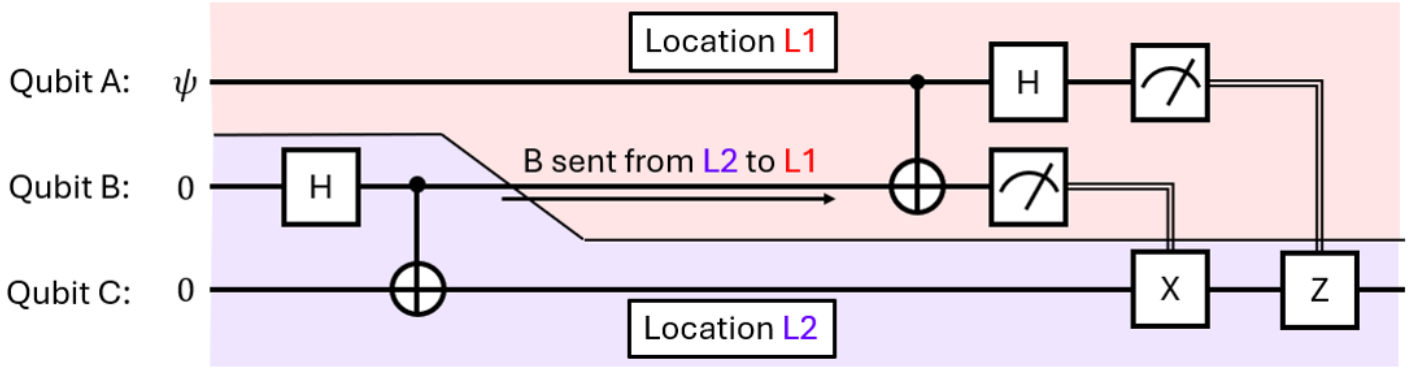
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1 - Not Relevant	2	3	4 - Extremely Relevant

Please provide any comments or suggestions regarding the wording of the item above.

Question

What is one non-local resource used in the following quantum teleportation circuit?



A non-local Bell state

A non-local quantum gate

A non-local quantum measurement

There are no non-local resources required

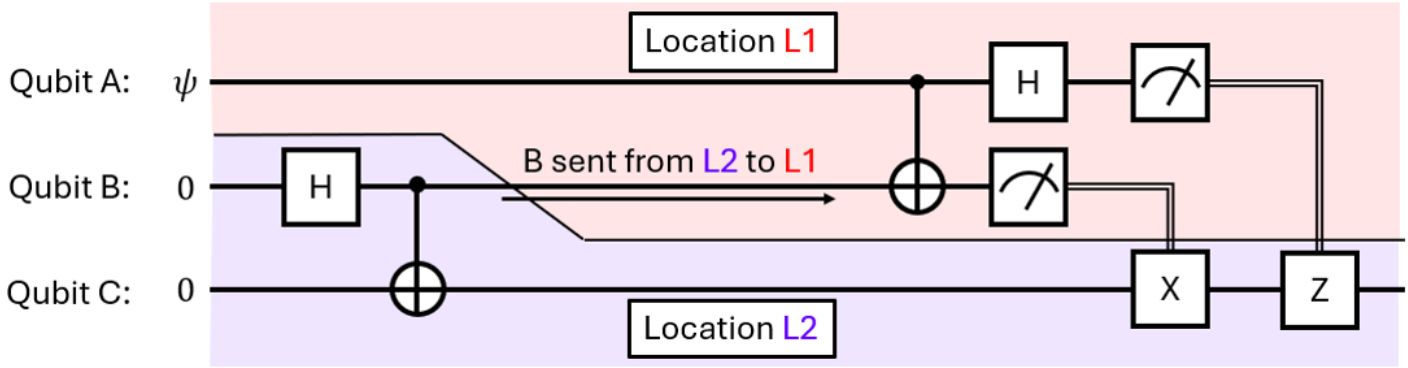
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Question

How can the following quantum teleportation circuit be modified to copy the initial state of qubit A to qubit C while maintaining the state at qubit A?



Remove the measurement on qubit A

Repeat the protocol in reverse

Replace the measurements with non-local quantum gates

Such a modification is not possible

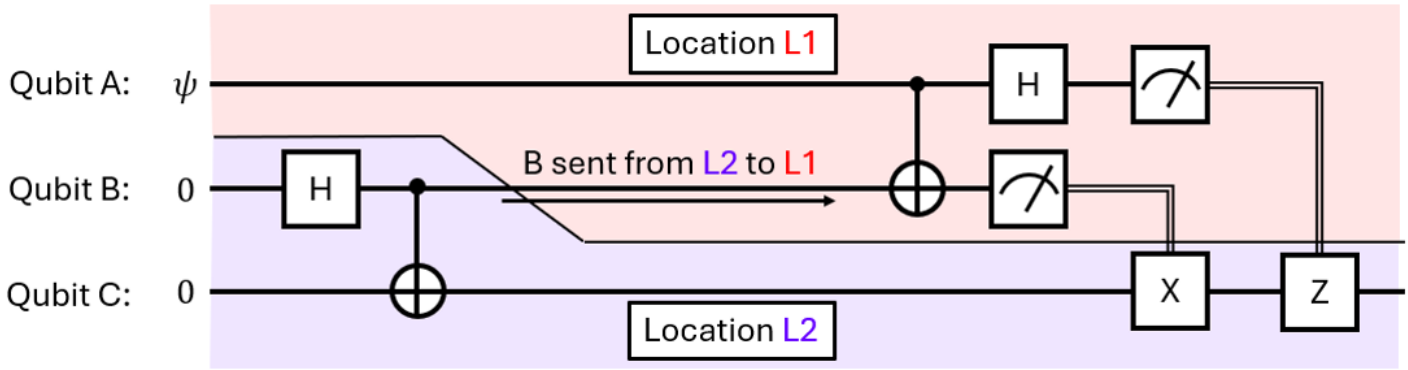
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Question

What quantum information does the following quantum teleportation circuit transmit?



The initial state of qubit A

The initial state of qubit B

The initial state of qubit C

The Bell state of qubits B and C

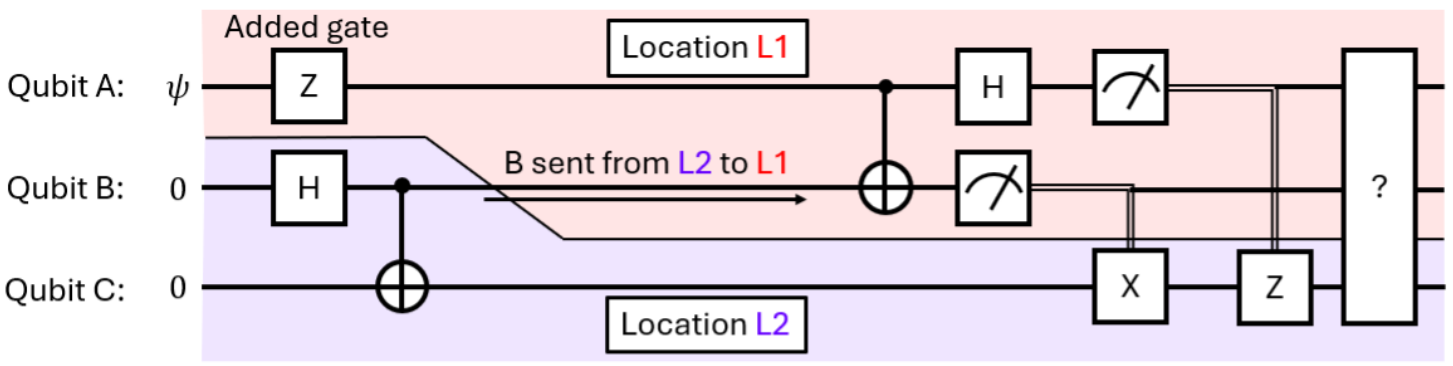
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Question

The canonical quantum teleportation circuit is modified by adding an initial Z gate to qubit A, as shown here. What common single-qubit gate should be added at the end of the circuit to invert the added Z gate?



A Z gate applied to qubit A

A Z gate applied to qubit B

A Z gate applied to qubit C

No gate needs to be added

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Decoherence

Decoherence

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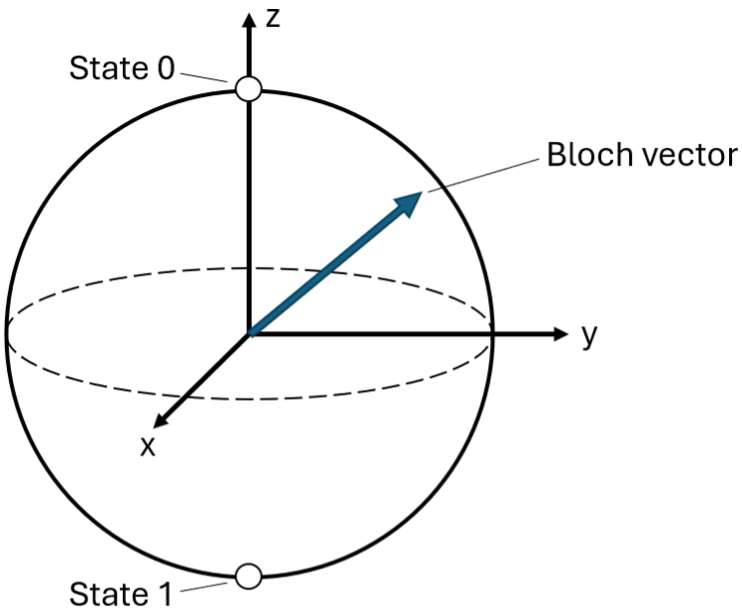
Glossary of Quantum Information Science Concepts

As an emerging field, quantum information science has a rapidly developing vocabulary. In this test, we use specific definitions that capture key concepts. More-general definitions may be encountered in advanced classes. The following glossary presents the specific definitions used through this test but does not contain all the information required to answer the questions.

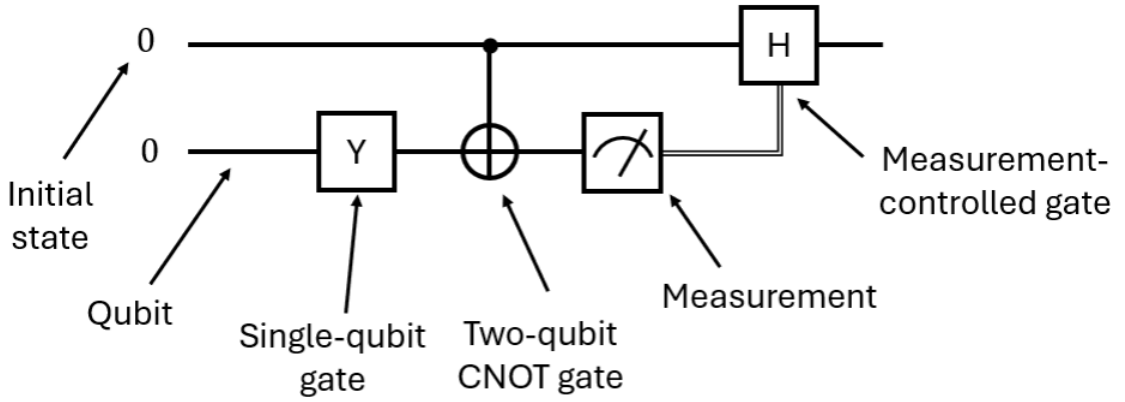
Quantum Bit (a.k.a. **Qubit**): The basic unit of quantum information. It is the quantum analogue of a classical bit, with two orthogonal basis states 0 and 1.

Superposition: The ability of a quantum system to be in multiple basis states at the same time.

Bloch Vector: A method of visualizing the state of a single qubit by converting information about its quantum state to a point on or within a sphere. For more than one qubit, the Bloch vector representation does not always provide complete information. A picture of a Bloch vector is shown here.



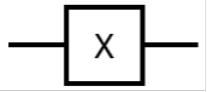
Quantum Circuit: A sequence of quantum operations applied to a system of qubits. A picture of a quantum circuit is shown here. The individual concepts (measurements, gates, etc.) are defined below.



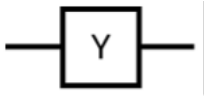
Measurement: The act of examining the state of the system, which causes the system to collapse probabilistically to one of the basis states (0 or 1) of the measured qubit. *For this test, all measurements are assumed to be projective measurements of a single qubit in the computational basis.*

Quantum Gate: A quantum operation that can change the state of one or more qubits via a generalized rotation. *For this test, the term gate will be used exclusively for ideal unitary quantum gates.*

X Gate: A quantum gate that rotates the Bloch vector of a single qubit by 180 degrees about the X axis.



Y Gate: A quantum gate that rotates the Bloch vector of a single qubit by 180 degrees about the Y axis.



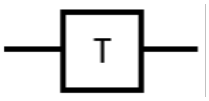
Z Gate: A quantum gate that rotates the Bloch vector of a single qubit by 180 degrees about the Z axis.



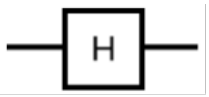
S Gate: A quantum gate that rotates the Bloch vector of a single qubit by 90 degrees about the Z axis.



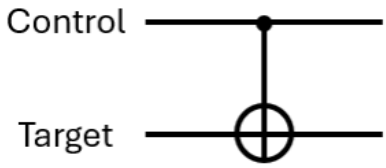
T Gate: A quantum gate that rotates the Bloch vector of a single qubit by 45 degrees about the Z axis.



Hadamard (H) Gate: A quantum gate that rotates the Bloch vector of a single qubit by 180 degrees about the X+Z axis.



Controlled Not (CNOT) Gate: A two-qubit quantum gate that performs an X gate on the target qubit if and only if the control qubit is in state 1.

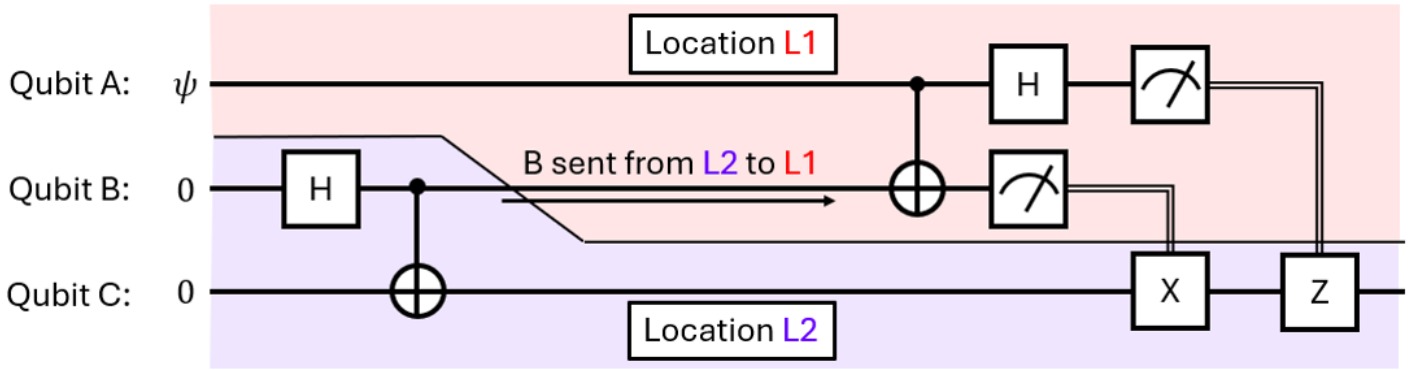


Entanglement: A fundamental quantum property in which the quantum state is shared among two or more qubits and not a property of any single qubit. When qubits are

entangled, measurement of one qubit is influenced by measurements of the other qubits.

Bell State: A specific two-qubit state that is maximally entangled. The four Bell states form a useful basis for describing two-qubit states.

Quantum Teleportation: A procedure for transmitting quantum information from one location to another using entanglement. A picture of a canonical teleportation circuit is shown here.



Decoherence: A process in which a quantum system's state loses information due to interactions with its surrounding environment. *For this test, all decoherence is assumed to be of Lindblad form (non-unitary, Markovian) such that dynamic decoupling does not protect against it. Noise from imperfect gates is considered as a form of decoherence.*

Provide any additional comments or suggestions regarding the wording of the glossary here.

Question

Two qubits start in an unknown state, after which they are subject to decoherence that

changes the state of the system. Is it possible to reverse the decoherence and return the qubits to their original unknown state?

Yes, it is possible to reverse the decoherence

No, it is not possible to reverse the decoherence

It is only possible if the qubits were originally entangled

It is only possible if the qubits were not originally entangled

Please rate the item above based on content relevance.

1 - Not Relevant	2	3	4 - Extremely Relevant

Please provide any comments or suggestions regarding the wording of the item above.

Question

Two qubits start in an unknown state. The first qubit is measured, which returns state 0. After some time, in which no unitary gates are applied, the first qubit is measured again, which returns state 1. What is the most likely explanation of this last result?

Decoherence due to the environment

Entanglement with the second qubit

Quantum teleportation of the second qubit's state

The first qubit was in a superposition

Please rate the item above based on content relevance.

1 - Not Relevant 2 3 4 - Extremely Relevant

Please provide any comments or suggestions regarding the wording of the item above.

Question #22 (Decoherence)

Quantum error correction is commonly used to address what issue?

Decoherence from the environment
End-of-circuit measurements
Entanglement created by two-qubit gates
Single-qubit gates

Please rate the item above based on content relevance.

1 - Not Relevant 2 3 4 - Extremely Relevant

Please provide any comments or suggestions regarding the wording of the item above.

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